

Hardware Components Identification -- Guided Notes

Unit 2.1 | CompTIA Network+ N10-009 Obj. Core 1 3.4 | N10-008

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

LGA (Land Grid Array) sockets have _____ on the motherboard socket itself, while the CPU has flat _____. PGA (Pin Grid Array) sockets have pins on the _____ and the socket has _____. Intel uses _____ sockets; AMD traditionally used _____ sockets (though AM5 is LGA).

Question 2

On a motherboard, a PCIe x16 slot is _____ than a PCIe x1 slot. An x16 slot is used primarily for _____. A PCIe x1 slot is used for smaller cards such as _____ and _____. A graphics card designed for an x16 slot will/will not fit in an x1 slot.

Question 3

An M.2 slot on a motherboard is a small _____ -shaped connector on the board surface. It accepts an M.2 SSD that mounts flat against the board and is secured with a single _____. A SATA port, by contrast, is an _____ -shaped connector used with a cable to connect 2.5-inch and 3.5-inch drives.

Question 4

The 24-pin ATX connector supplies _____ from the power supply to the motherboard. The small headers at the front edge of the motherboard connect to _____ panel cables. These cables control the _____ button, _____ LED, and _____ LED.

Question 5

SO-DIMM (Small Outline DIMM) RAM modules are used in _____ and are physically _____ than full-size DIMMs used in desktops. A standard desktop DIMM is approximately _____ mm long, while a SO-DIMM is approximately _____ mm long.

Question 6

A system with _____ graphics uses a GPU built into the CPU or motherboard chipset. Display output ports in this case are located on the _____. A _____ GPU is a separate add-in card. When a discrete GPU is installed, display cables should be connected to the _____, not the motherboard.

Question 7

High-end discrete GPUs require supplemental power from the PSU via _____ connectors (6-pin or 8-pin PCIe power connectors). These connectors supply the _____ V rail directly to the GPU. If these connectors are not plugged in, the system may _____ or the GPU may not be detected.

Question 8 -- Real World Application

REAL WORLD: A student opens a used desktop tower to identify its components before upgrading it. They see a short rectangular connector on the surface of the motherboard, a long connector on the motherboard near a large fan, two different-length slots on the motherboard, and a cable going to the drive labeled SATA. Identify each component and explain the function of two of them in detail.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-1-hardware-components-identification/2-1-hardware-components-identification-answer-key.pdf
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